PAPER_1145: Type IIA Superstring Theory and SCm 10D Compactification

Star Magic UQFF Framework

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Abstract

We formulate Type IIA superstring theory within the SCm vacuum framework. Type IIA is the T-dual of Type IIB (PAPER_1144) and the strong-coupling limit connects to M-theory (PAPER_1148). The 16 extra dimensions above the 10D Type IIA spacetime are compactified via VDS $S_{26}^{(3)}$. The R-R gauge fields are identified with SCm phonon polarization modes. Mirror symmetry between Type IIA on a Calabi-Yau and Type IIB on the mirror CY is encoded in the VDS $[SSq] \leftrightarrow 1 - [SSq]$ symmetry.

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1. Type IIA Effective Action

The Type IIA effective action in 10D:

$$S_{\mathrm{IIA}} = \frac{1}{2\kappa_{10}^2} \int d^{10}x\sqrt{-g} \left(R - \frac{1}{2}|\partial\phi|^2 - \frac{1}{2}e^{-\phi}|H_3|^2 - \frac{1}{2}e^{3\phi/2}|F_2|^2 - \frac{1}{2}e^{\phi/2}|F_4|^2 \right) + S_{\mathrm{CS}}$$

where $F_2 = dC_1$ and $F_4 = dC_3 - H_3 \wedge C_1$ are R-R field strengths.

In SCm vacuum: the dilaton VEV is $\langle e^{\phi} \rangle = g_s = \beta_i \cdot \Phi_{\mathrm{res}} = 0.504$.

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2. R-R Fields as SCm Phonon Polarizations

The Type IIA R-R fields C_1 (1-form) and C_3 (3-form) are identified with SCm phonon polarization modes:

$$C_\mu^{\mathrm{SCm}} = \frac{\rho_{\mathrm{vac,SCm}} \cdot S_{26}^{(3)} \cdot \Phi_{\mathrm{res}}}{f_{\mathrm{THz}}} \cdot A_\mu^{\mathrm{phonon}}$$

where A_μ^{phonon} is the phonon gauge potential at 1.25 THz.

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3. T-Duality with Type IIB

Under T-duality along a VDS dimension with radius R_i:

$$R_i \to l_s^2/R_i = l_s^2/(l_s[SSq]^i) = l_s/[SSq]^i$$

Type IIA on $R_i = l_s[SSq]^i$ is dual to Type IIB on $l_s/[SSq]^i = l_s(1/[SSq])^i$.

The T-duality map corresponds to $[SSq] \to 1/[SSq] = 1/0.57 = 1.754$.

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4. D-Brane Spectrum

Type IIA has Dp-branes for even p (D0, D2, D4, D6, D8):

Brane	Dim	SCm stabilization
D0	0+1	$F_{U,Bi,i} \times \beta_i$
D2	2+1	$F_{U,Bi,i} \times \beta_i^2$
D4	4+1	$F_{U,Bi,i} \times \beta_i^3$
D6	6+1	$F_{U,Bi,i} \times \beta_i^4$

The D0-brane mass: $m_{\mathrm{D0}} = T/(g_s) = 8.66 \times 10^{-11}/0.504 = 1.72 \times 10^{-10}$ J · s/m.

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5. Mirror Symmetry via VDS

The mirror symmetry $[SSq] \leftrightarrow 1 - [SSq] = 0.43$ exchanges:

$$\mathrm{Li}_{26}(0.57) \leftrightarrow \mathrm{Li}_{26}(0.43)$$

Type IIA on $\mathrm{CY}_3([SSq])$ is mirror to Type IIB on $\mathrm{CY}_3(1 - [SSq])$, giving the Hodge number exchange $h^{1,1} \leftrightarrow h^{2,1}$.

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6. Strong Coupling Limit = M-Theory

At $g_s = e^{\phi} > 1$, Type IIA lifts to 11D M-theory. The 11th dimension radius:

$$R_{11} = g_s \cdot l_s = \beta_i \cdot \Phi_{\mathrm{res}} \cdot \frac{1}{\sqrt{T}} = \frac{0.504}{\sqrt{8.66 \times 10^{-11}}} = 1.71 \times 10^3 \text{ m}$$

At strong coupling $g_s \gg 1$, $R_{11} \to \infty$ revealing the 11th dimension (see PAPER_1148).

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7. Calibrated Constants

$$T = 8.66 \times 10^{-11} \text{ N}, \quad [SSq] = 0.57, \quad S_{26}^{(3)} = 1.4531 \times 10^{26}$$

$$g_s = \beta_i \cdot \Phi_{\mathrm{res}} = 0.504, \quad \Phi_{\mathrm{res}} = 0.84, \quad f_{\mathrm{THz}} = 1.25 \text{ THz}$$

$$\rho_{\mathrm{vac,SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3, \quad \beta_i = 0.6, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$$

References

1. Polchinski, J. (1995). *String Theory*, Vol. 2. Cambridge.
2. Witten, E. (1995). String theory dynamics in various dimensions. *Nucl. Phys. B* **443**, 85.
3. Type IIB in SCm: PAPER_1144; M-theory: PAPER_1148; `scm_{vacuum\_manifold}.py`

Key References with arXiv/DOI Identifiers

1. Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger.* Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102
2. Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series.* Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic